

Question:

An isosceles right triangle with legs of length \$8\$ is partitioned into \$16\$ congruent triangles as shown. The shaded area is

$\text{(A)}\ 10 \quad \text{(B)}\ 20 \quad \text{(C)}\ 32 \quad \text{(D)}\ 40 \quad \text{(E)}\ 64$

Reasoning + Answer:

<think>

1. **Understanding the Problem :**

- We are given an isosceles right triangle with legs of length 8 .
- This triangle is partitioned into 16 congruent smaller triangles .
- We need to find the shaded area .

2. **Area of the Large Triangle :**

- The formula for the area of a right triangle is $\left(\frac{1}{2} \times \text{base} \times \text{height}\right)$.
- Here , the base and height are both 8 .
- Therefore , the area of the large triangle is :

$$\begin{aligned} \text{Area}_{\text{large}} &= \frac{1}{2} \times 8 \times 8 = 32 \end{aligned}$$

3. **Area of Each Smaller Triangle :**

- The large triangle is divided into 16 smaller congruent triangles .
- Therefore , the area of each smaller triangle is :

$$\text{Area}_{\text{small}} = \frac{\text{Area}_{\text{large}}}{16} = \frac{32}{16} = 2$$

4. **Shaded Area Calculation :**

- The shaded area consists of 10 of these smaller triangles .
- Therefore , the shaded area is :

$$\text{Shaded Area} = 10 \times \text{Area}_{\text{small}} = 10 \times 2 = 20$$

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<answer> 20 </answer>